

EX. NO : 8

Simple Perceptron Implementation for AND gate

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Aim:

To implement a simple perceptron algorithm to learn the AND logic gate.

Procedure:

1. Initialize weights and bias to zero.
2. For each training example, calculate weighted sum and apply step activation function.
3. Calculate error as difference between expected and predicted output.
4. Update weights and bias using the perceptron learning rule.
5. Repeat for several epochs until convergence.
6. Test the trained perceptron on all input combinations.

Program Coding:

```
# Training data: inputs and labels
```

```
training_inputs = [
```

```
    [0, 0],
```

```
    [0, 1],
```

```
    [1, 0],
```

```
    [1, 1]
```

```
]
```

```
labels = [0, 0, 0, 1]
```

```
# Initialize weights and bias
```

```
weights = [0.0, 0.0]
```

```
bias = 0.0
```

```
# Learning rate
```

```

lr = 0.1

# Activation function (step function)

def step(x):

    return 1 if x >= 0 else 0

# Training loop

for epoch in range(10): # Train for 10 epochs

    print(f'Epoch {epoch+1}')

    for inputs, label in zip(training_inputs, labels):

        # Weighted sum

        linear_output = sum(w * inp for w, inp in zip(weights, inputs)) + bias

        prediction = step(linear_output)

        # Update weights and bias

        error = label - prediction

        for i in range(len(weights)):

            weights[i] += lr * error * inputs[i]

        bias += lr * error

        print(f' Input: {inputs}, Prediction: {prediction}, Error: {error}')

    print(f' Weights: {weights}, Bias: {bias}\n')

# Test perceptron after training

print("Testing trained perceptron:")

for inputs in training_inputs:

    linear_output = sum(w * inp for w, inp in zip(weights, inputs)) + bias

    prediction = step(linear_output)

    print(f'Input: {inputs}, Output: {prediction}')

```

Output:

Epoch 1

Input: [0, 0], Prediction: 1, Error: -1

Input: [0, 1], Prediction: 0, Error: 0

Input: [1, 0], Prediction: 0, Error: 0

Input: [1, 1], Prediction: 0, Error: 1

Weights: [0.1, 0.0], Bias: 0.0

...Testing trained perceptron:

Input: [0, 0], Output: 0

Input: [0, 1], Output: 0

Input: [1, 0], Output: 0

Input: [1, 1], Output: 1

Result:

The perceptron successfully learned the AND gate logic function by adjusting weights and bias to correctly classify all inputs.